

1 Rollout

$$\pi_t$$



GAE

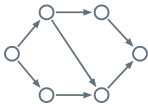
$$\hat{A}_r, \hat{A}_{c_k}$$

episode cost

$$\bar{J}_k$$

2 Flow inputs

clustered states



edge attributes



3 Constraints

soft budgets

$$\langle C_k, F \rangle \leq b_k$$

$$b_k = B_k / L$$

hard exclusions



$$F_H = 0$$

4 Entropic projection

$$\min_{F \geq 0} \left[-\langle A, F \rangle + \varepsilon \text{KL}(F \| \hat{F}) \right]$$

s.t. flow balance + constraints

$$\hat{F}$$

$$F^*$$



Target flow is not executed

advantages

$$\bar{\lambda}_k^* \leftarrow \text{clip}_c + \text{EMA} \leftarrow \lambda_k^*$$

budget duals

5 Priced policy update

$$\beta_k \leftarrow [\beta_k + \eta(\bar{J}_k - B_k)]_+$$



$$\lambda_k$$

$$\tilde{A} = \frac{\hat{A}_r - \sum_k \lambda_k \hat{A}_{c_k}}{1 + \sum_k \lambda_k}$$

$$\rightarrow \text{PPO/TRPO} \rightarrow \pi_{t+1}$$

next rollout